

Where Cognition Lives: Dissecting Emergent from Computed Function in a Minimal Complete Cognitive Architecture

Anonymous authors

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Abstract

A cognitive architecture is more than the module that reasons. It must also decide how long to think, what deserves the effort, and when looking is better than planning. We built a minimal but complete system—a recurrent reasoner with adaptive halting, a homeostatic control field, and a value module—and spent a research program asking one question of every part: does this function *emerge* from gradient descent, or must it be *computed* by explicit machinery? The answers are sharp. Competence emerges. Stopping emerges too, and appears to be worth more than everything decidable in advance—but that appearance is instrumentation: in our information hierarchy, payoff at closely matched mean compute climbs from 0.467 (uniform allocation) through 0.546 (difficulty) to 0.698 (ex-ante value), which matches the class-stake oracle to three decimals—an equality that is partly by construction, since our stake sensor is perfect. The further climb to 0.921 (posterior self-observation) does *not* survive audit: PonderNet-style halting returns a halting-weighted *mixture* of hidden states while every forced-depth baseline returns a single state, and the language head is trained on the mixture alone. Equalizing the readout annihilates the apparent advantage of native execution (residual +0.000 [0.000, 0.000]), and with the readout held fixed and the budget matched, knowing which instance needs how much compute is worth +0.0011 [+0.0003, +0.0019]. Adding value on top of the posterior likewise buys nothing (+0.0002 ± 0.0004). Value does *not* emerge—trained couplings capture zero of an available payoff that an explicit allocator captures completely (+0.151, routing correlation +0.79)—while anticipation has no payoff to capture at all in these families ($\leq +0.001$ across 31 configurations), so the second-order decisions that do pay must be computed, at least where value is orthogonal to content as it is here by construction. On a frozen LLM actuator the same instruments show that the standard test-time lever, self-consistency voting, is a measured bound (+0.0236 [+0.0150, +0.0326]) and that inter-sample agreement is nearly worthless as a stopping signal—its mass concentrates on wrong answers—so watching oneself think is nearly worthless in *both* regimes, a convergence that was invisible while one of the two was measured through a readout the other did not share. Almost every negative result in the program carries its mechanism—the exceptions are declared as unadjudicated—and the protocol that produced them—preregistration, adversarial panels, kill-gates, replication rules, and positive controls for every null we assert—is part of the contribution. We close by executing our own falsifiable prediction. In a cliff-cost family, where the compute an instance needs is invisible until it is spent, value under commitment pays +0.1312 [+0.1124, +0.1502], roughly seven times the point estimate (+0.019) in a smooth family—yet not because the cliff shifts information toward ex-ante decisions (the ex-ante fraction of the attainable range is 0.759 versus 0.763; difference −0.004 [−0.150, +0.241], a post-hoc observation compatible with equality but imprecisely estimated), but because it multiplies the attainable range fivefold ($5.1 \times [3.4, 8.2]$). On this evidence the magnitude of an allocation problem and the structure of its information behave as separate axes—the first measured decisively, the second only bounded.

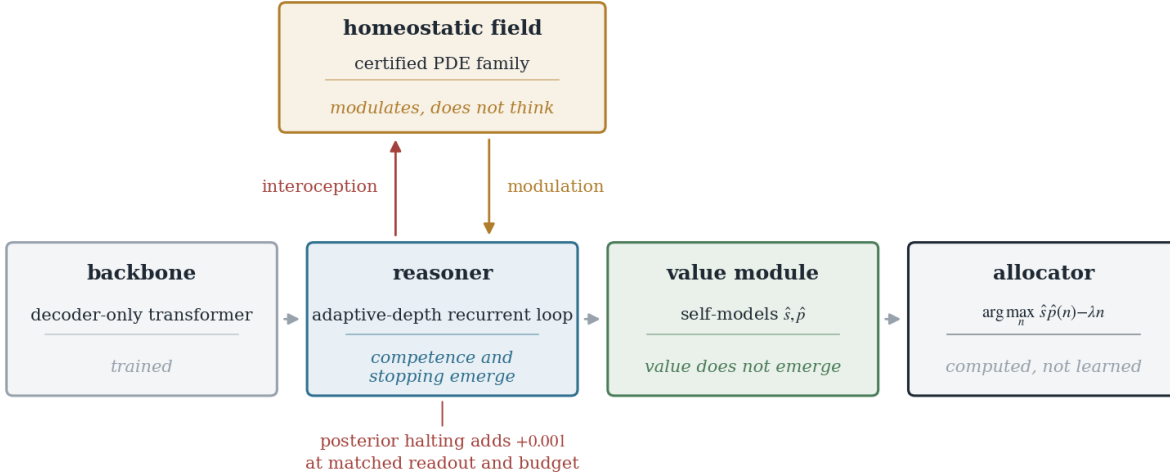


Figure 1: The minimal complete architecture, annotated with the program’s verdict for each component. Competence and stopping emerge in the reasoner; the certified homeostatic field modulates compute without carrying content; value-shaped decisions do not emerge from any trained coupling and are computed by an explicit allocator over learned self-models. The posterior channel—the halting head observing the evolving state—*appears* worth +0.223 over everything decidable ex-ante, but a readout audit attributes that gap to how adaptive-depth arms are read rather than to what they know: at matched readout and budget it is worth +0.0011 (§5).

1 Introduction

Two intuitions about machine cognition pull in opposite directions. One says that everything worth having emerges from scale and gradient descent (Wei et al., 2022a)—and that architectural commitments mostly get in the way—though what truly emerges is itself contested (Schaeffer et al., 2023). The other says that cognition is an engineered stack—perception here, memory there, an executive on top (Anderson et al., 2004; Laird, 2012; LeCun, 2022)—and that each function needs its dedicated box. Neither intuition survives contact with measurement. The purpose of this paper is to put the boundary itself under the instrument: for each function of a small but complete cognitive architecture, to determine experimentally whether it emerges from training or has to be computed by explicit machinery, and to report the mechanism behind every answer.

Our testbed is deliberately modest in scale and deliberately complete in structure (Fig. 1). A decoder-only transformer backbone feeds a recurrent *reasoner* with adaptive halting in the style of PonderNet (Banino et al., 2021; Graves, 2016; Dehghani et al., 2019); a low-dimensional homeostatic field—the subject of a companion paper (Anonymous, 2026)—modulates its compute; an explicit slot memory and a value module complete the system.

Tasks are algorithmic (S_5 permutation composition and relatives), chosen because their difficulty and information structure can be controlled exactly, and because fixed-depth transformers provably struggle with them and shortcut solutions fail to length-generalize (Merrill & Sabharwal, 2023; Liu et al., 2023; Anil et al., 2022; Zhou et al., 2024). In the final chapters we swap the trained substrate for a frozen 14B-parameter language model (Qwen Team, 2024) acting as a text actuator, which lets us ask the same questions of test-time compute in the LLM regime (Wei et al., 2022b; Wang et al., 2023; Snell et al., 2024).

The paper makes five contributions. **(1)** An *information hierarchy* for compute allocation, measured at matched compute on frozen solvers: uniform 0.467 < difficulty 0.546 < ex-ante value 0.698, which is the hierarchy’s honest ceiling. The apparent fourth rung, posterior self-observation at 0.921, is *not* commensurable with the three below it—it is read from a halting-weighted mixture of states that no ex-ante arm

shares—and at matched readout and matched budget, knowing which instance needs how much compute is worth $+0.0011$ [$+0.0003$, $+0.0019$] (§5). **(2)** The *computed-decision thesis*: trained couplings leave the entire value-shaped payoff on the table while an explicit argmax allocator over learned self-models captures all of what the stake-class oracle can reach—an equality that is partly by construction, since our stake sensor is perfect (§4). **(3)** The *precondition of governance* on LLM actuators: before asking how to govern test-time compute one must measure whether the compute lever has dynamic range at all, and for self-consistency voting on our actuator it barely does—what any stake-aware allocation can buy is a two-sided bound, above zero and below our relevance threshold, with a mechanism (§6). **(4)** A *negative space* of twelve dead or suspended hypotheses (Table 1), each retired with its mechanism rather than a bare null (§7), produced by a protocol we argue is a contribution in itself (§8). **(5)** The *cliff result*: executing our own preregistered prediction about all-or-nothing cost landscapes, with an outcome that confirms the headline hypothesis, fails to reproduce the strong form of the prediction (with an interval too wide to refute it either), and fails its own second preregistered hypothesis in its preregistered form (§9).

Common experimental setup. Throughout §2–§5, instances are length- K composition problems over S_5 (compose K group generators and report the resulting permutation; out-of-distribution means K beyond the training range), solved by decoder-only transformers of 4–6M parameters (per-variant counts in the repository). A *solver* is a trained checkpoint evaluated frozen (forward-only); the twelve solvers of §4–§5 are six seeds $\times$ two independent runs, our standing replication unit. Each instance carries a *stake* $s \in \{1, 8\}$ with $P(s=8) = 0.15$, assigned independently of content and visible only as metadata; *payoff* is stake-weighted accuracy, $\sum_i s_i c_i / \sum_i s_i$, on a common frozen evaluation set of 16k instances. An *arm*—in the bandit sense (Gittins, 1979; Lattimore & Szepesvári, 2020)—is an allocation policy mapping instances to reasoner iterations under a shared compute budget, compared at matched mean iterations. A *kill-gate* is a cheap preregistered go/no-go measurement empowered to cancel an expensive experiment; design calibration was limited to at most three predeclared adjustment rounds per task family, after which the design froze.

Relation to prior work. Our questions descend from rational metareasoning: the hierarchy of §5 is an empirical measurement, at matched compute, of the value of computation in the sense of Russell & Wefald (1991)—a lineage running through Horvitz (1987) and Hay et al. (2012)—and its resource-rational reading follows Lieder & Griffiths (2020). Adaptive-depth and conditional-compute architectures (Graves, 2016; Banino et al., 2021; Dehghani et al., 2019; Elbayad et al., 2020; Schwartz et al., 2020; Raposo et al., 2024) supply our stopping machinery; we measure *which information* makes stopping and allocation pay, not only how to learn them. The test-time-compute literature—voting, verifiers, search, and repeated sampling (Wei et al., 2022b; Wang et al., 2023; Cobbe et al., 2021; Lightman et al., 2024; Yao et al., 2023; Madaan et al., 2023; Brown et al., 2024; Snell et al., 2024)—supplies the levers of §6–§9; our contribution there is the precondition—measuring a lever’s dynamic range, and the reliability of self-agreement (Kadavath et al., 2022; Lin et al., 2022), before optimizing over it. The mechanistic facts we lean on, induction heads and grokking, are from Elhage et al. (2021); Olsson et al. (2022); Power et al. (2022); Nanda et al. (2023).

2 The substrate and its certification

Before asking where cognition lives one must be sure there is cognition in the building. We resist defining “reasoning” by decree. In this program a *reasoner* is not a module but a *regime* of a (task, system, training) triple, certified by a battery of four measurements we call T-M-I-P: the **T**ask demands serial computation; the **M**emory of the trajectory carries content; halting is **I**nformed; and de**P**th extrapolates. The load-bearing test is **M**. If the recurrent state actually carries the computation, then transplanting the state mid-trajectory between two instances must transplant the answer. It does: in state-swap interventions the prediction follows the donor state in 0.83–0.92 of cases (the *cycle* family, which cleared the seriality gate in two of three seeds; the *adjacent* family failed it outright here and carries no cognitive claim in this paper, although it remains a valid compute-allocation benchmark and is, as it happens, the family in which the order effect of §3 survives—a tension we flag rather than resolve) and never spontaneously reverts to the recipient’s answer (0.00), and representations crystallize over ticks (readout fidelity $0.5 \rightarrow 0.85$ – 0.89 across seeds). Informed stopping exists as well (AUC 0.79–0.90), though certification was not linear: the battery initially *failed* in all twelve of its preregistered cells against its preregistered thresholds, an earlier inverted-AUC measurement proved to be a

ceiling artifact, and the passing numbers come from preassigned remedies and a re-certification—an ordering we report as part of the audit trail.

The battery also earned its keep negatively. An apparent side effect of integrating the control field—a small apparent accuracy cost (“the touch”) of -0.055 , $p = 0.011$ —was not confirmed by a preregistered replication with fresh contemporary controls: the replicated effect was -0.011 $[-0.068, +0.045]$ —an interval compatible both with zero and with the original effect—so the finding rests on evidence no stronger than single-run-per-cell noise ($\sigma_{\text{run}} \approx 0.04\text{--}0.05$), and we retired it. Out of that episode came a standing rule of the program: effects below ~ 0.05 require at least two independent runs per cell, because seed-pairing does not pair run noise. We will meet this rule again in §8.

3 The homeostatic governor: substance no, structure only in part, certifiability yes

The architecture’s most speculative component is a low-dimensional homeostatic field defined on the graph of the model’s modules and driven by a family of partial differential equations on the graph Laplacian—damped wave (forced Klein–Gordon), its diffusive limit, and a KdV-type dispersion—inspired by the brain’s neuromodulatory systems (Barrett, 2017; Man et al., 2022; Vecoven et al., 2020). The full formalism, its stability certificates (an antisymmetric-placement lemma with a flutter threshold $\beta\rho(A^3) < 2\zeta\omega_0^2$ that is exact when the stiffness is a multiple of the identity and a design guide otherwise, discrete Schur–Cohn certificates, an unconditionally stable IMEX integrator), and the complete experimental record are in the companion paper (Anonymous, 2026). Here we need its conclusion, because the rest of the program builds on it: *substance no, structure only in part, certifiability yes—the field modulates, it does not think, and a learned governor of matched interface modulates just as well.*

Substance, no: the *type* of physics is irrelevant to accuracy. Wave, diffusion, gated mixtures, nonlocal Poisson coupling, and even a 2D Navier–Stokes flow substrate all reach the same accuracy, for a measured reason: the training gradient coarse-grains time, laminating every demand for modulation into a quasi-static set-point; and an incompressible flow cannot concentrate information at a point ($\nabla \cdot u = 0$ conserves area). Making the field the *source* of computation is catastrophic ($d_z = 4.06$ worse than a GRU of matched interface). Structure, in part: in one of the two generator families—and, once capacity is equalized, only there—the *second-order* character of the dynamics—inertia—confers on out-of-distribution compute *allocation* a robustness that the non-stateful learned halting control lacks (the contrast against the field-free control below does not by itself isolate order from the presence of a field): an initial exploratory signal (correlation 0.63 versus 0.20 on the original task) that replicated under the companion’s preregistered v3 protocol at smaller magnitude (0.143 vs. 0.002 and 0.268 vs. 0.213 across the two generator sets; paired $p=0.025/0.028$, directionally consistent in both but not surviving Holm-4 individually) and survived an A/B against a numerics bug that had frozen the physical constants. A reviewer-requested deconfounding campaign (twenty fresh seeds; first order with equalized caps; a matched-interface GRU replacing the integrator) then delivered the chapter’s final, humbler form: the order effect is strong in the **adjacent** generator family ($+0.087$, $[+0.042, +0.132]$, $t=4.0$, $p=3.5\times 10^{-4}$) and is not detected in **cycle_transp** once capacity is equalized ($+0.014$, $[-0.013, +0.040]$, n.s.), so under Holm-2 the order hypothesis is not confirmed as a family-level claim; and the learned GRU governor of matched interface ties the field in **adjacent** ($+0.006$, $[-0.051, +0.062]$, n.s.) and is nominally *better* in **cycle_transp** (-0.035 , $[-0.067, -0.002]$), with mutual non-inferiority of long-stratum accuracy at a 0.02 margin. What survives is the value of a stateful modulating governor with this interoceptive interface—of which the field is the *certifiable* implementation—rather than any superiority of the physics.

The field’s final audit—the evidence-accumulator kill-gate—is reported in full in the companion; here we need its verdict. The last live hypothesis for making the field load-bearing was that it could serve as a temporal *evidence accumulator*—a damped second-order filter integrating the reasoner’s noisy per-tick posterior stream, the physical form of sequential analysis. A kill-gate whose primary cell and 0.03 pass threshold were fixed in a dated design document before the verdict (not a formal preregistration; the probe was recalibrated after a first null) tested the premise on twelve frozen solvers with a certified probe (its positive control detected planted distributed signals at the threshold scale): does the posterior stream carry information about success beyond the last tick? It does not: $\Delta\text{AUC} = +0.0007$ $[-0.0065, +0.0079]$ in the predeclared primary cell, with every secondary cell within $[-0.0012, +0.0073]$ —none near the 0.03 threshold.

The mechanism is structural. The recurrent state already integrates its own history; the last tick is a sufficient statistic, and the accumulator’s niche is occupied by construction. With that, every employment imagined for the field is measured, and none is performed better by the field than by a dedicated component—not even compute control, where a matched-interface GRU ties it in one generator family and is nominally better in the other. What is left to the field is not a performance niche but a licensing one: it is the only implementation of this governor that ships with a placement dichotomy, a flutter threshold and an unconditional IMEX bound.

4 Second-order decisions do not emerge; they are computed

The center of the program is a series of integration experiments asking where decisions about *what matters* and *how much to spend* come from. (We call these *second-order decisions*—decisions about the reasoning process, not within it; no relation to the second-order *dynamics* of §3.) The design rule throughout—learned the hard way—is to force capability through the *environment*, never through the loss.

Anticipation has no exploitable value in a smooth family. We built session environments—sequences of instances whose hidden difficulty regime switches stochastically, under a hard shared compute budget—where anticipating the regime from history could in principle pay, and scanned 31 configurations of stakes, budgets, horizons, and regime dynamics against an oracle ladder (up to a perfect Bayes filter over the whole history). The ceiling for anticipation was $\leq +0.001$ payoff: the value profile is smooth and regimes mix faster than any anticipatory policy can exploit, so the reactive stationary policy is already within $+0.001$ of the perfect-Bayes ceiling in every configuration we scanned. No GPU was spent on a confirmatory experiment that could not pay.

Gradient coupling is deaf, and weighted losses self-defeat. We then handed the trained system every advantage: a value channel with a near-perfect sensor (a learned stake predictor reaching AUC = 0.95; 1.00 from raw embeddings) wired into the halting mechanism, and trained end-to-end with value at stake. The behavioral result: routing correlation -0.01 in every arm—the coupling captures nothing. The mechanism is legible, and we suspect general though we measured it only here, and worth stating: *training suppresses task-irrelevant information from the state* (a probe recovers stake from raw embeddings at 1.00 but from the trained state at only 0.79); the state serves the loss, and anything the loss does not need decays. The complementary failure is starker. Weighting the cross-entropy by stakes—making the loss “feel” consequences—makes the model *worse on the very instances it up-weights* (0.786 versus 0.931 for a stake-blind control; effective sample size 0.40), and the blind control is better by $+0.147$ overall. Consequences in the loss are not a teacher; they are a variance machine. (Both dissections are exploratory branches that the preregistration did not anticipate, and we report them as such; the preregistered primary contrast—value-coupled versus uncoupled halting—was an exact null, $\Delta = +0.0000$.)

The explicit allocator captures the entire ex-ante ceiling. The same payoff that gradient coupling captured at zero is then captured completely by a computed decision: $n^* = \arg \max_n \hat{s} \cdot \hat{p}(\text{success} | n) - \lambda n$, where $\hat{s}$ and $\hat{p}$ are self-models fitted from the system’s own probes, and the argmax runs at evaluation time on frozen solvers. Against a difficulty-only baseline the allocator gains $+0.151$ ($t=24.97$ against the preregistered relevance threshold $\delta_0=0.02$, not against zero, paired at the seed level ($n=6$, $df=5$), one-sided $p = 8.7 \times 10^{-7}$; all twelve replicas positive, hence all six seed-level means positive) with routing correlation $+0.79$, and it reaches the class-information oracle—an allocator given the true stake class—to three decimals: one hundred percent of the ceiling attainable from ex-ante information. That equality is partly by construction (the stake sensor is perfect here, so $\hat{s}$ coincides with the stake); the non-trivial learned content is the success model $\hat{p}$, whose removal—a hand rule without the learned self-model—leaves $+0.06$ on the table. This is the computed-decision thesis: in this setting—where value is orthogonal to content by construction—second-order decisions do not emerge from the gradient; they are computed, and computing them works at the limit of the information our probes expose. Whether the verdict survives when value is correlated with content is untested.

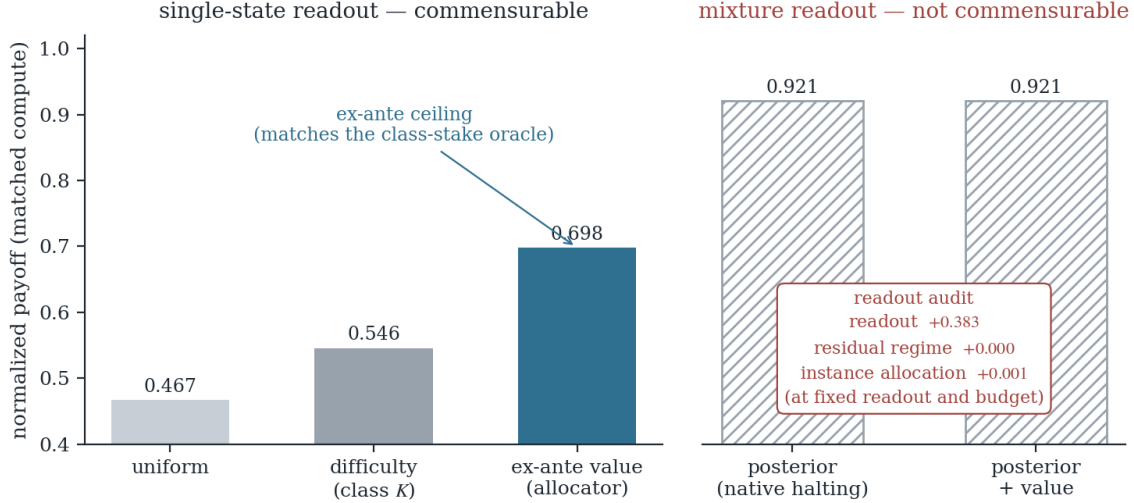


Figure 2: The information hierarchy on the trained substrate: normalized payoff at closely matched mean compute (ex-ante arms at $e=5$ exact, native arm at its emergent $\bar{n}=5.47$, i.e. 9% more compute; see the execution-regime caveat in §5), on twelve frozen solvers with a common 16k evaluation. The explicit ex-ante allocator reaches the class-stake oracle to three decimals, partly by construction. The top rung is *not* commensurable with the other three: it is read from a halting-weighted mixture of states while the ex-ante arms are read from a single state, and the readout audit of §5 attributes the whole apparent +0.223 to that difference (residual execution regime +0.000; instance allocation at fixed readout +0.001). Value on top of the posterior buys $+0.0002 \pm 0.0004$.

5 The information hierarchy

Putting the arms side by side at closely matched compute (exact at $e=5$ among the ex-ante arms; the native arm runs at its emergent $\bar{n}=5.47$) yields the program’s central figure (Fig. 2). Uniform allocation earns 0.467. Knowing task difficulty earns 0.546. Knowing ex-ante value—through the explicit allocator of §4—earns 0.698, which equals the class-oracle ceiling. Letting the system *observe itself think*—the native adaptive halting reading the evolving state and stopping on its own learned schedule—*appears* to earn 0.921, read, crucially, through a readout the other three arms do not share (next paragraph). And biasing that native stopping with value, at matched expected compute and with a preregistered sweep of bias strengths, adds $+0.0002 \pm 0.0004$ (mean $\pm$ standard *deviation* across the twelve solvers; the standard error is 0.0001): nothing.

The +0.223 does not survive audit, and the audit is the chapter’s sharpest result. A *yoked* control, run at reviewer request, forced execution at exactly the per-instance depths the native halting chose (rounded $\mathbb{E}[n_i]$) and recovered almost none of the native advantage: against uniform forced allocation the native arm gains +0.430 [+0.397, +0.463], of which transplanted depths accounted for only +0.047 [+0.029, +0.065]. We first read the remaining +0.383 as the value of elastic, on-policy execution. It is not. It is the readout.

The readout audit. `AdaptiveHalting` returns $\sum_n p_n x_n$, a halting-weighted *mixture* of hidden states; the forced path returns the single state reached after the quota; and the language head was trained with cross-entropy on the mixture, so it has never decoded a single state. Reconstructing every arm post hoc from the *same* recorded native trajectory separates the two (twelve solvers, common evaluation; the reconstruction reproduces the native arm exactly, and the post-hoc single state at $\hat{n}$ agrees with the independently re-executed forced arm on all twelve). The decomposition is exhaustive to machine precision: the readout accounts for +0.383 [+0.341, +0.424], the residual execution regime for +0.000 [0.000, 0.000], and transplanted depth information for +0.047 [+0.029, +0.065]. Forcing execution costs nothing: it reproduces the native trajectory tick by tick. Reading it differently costs everything. The same artifact explains the prede-

clared VG-N3d control, which had measured -0.34 for forced-depth execution and attributed it to the cost of commitment.

With the readout held fixed. The audit invalidates the native-versus-forced comparison but not the question behind it, which we then ask cleanly: keep every arm inside the mixture family, over the same recorded states, and vary only the weights. Giving each instance the halting distribution of a *different* instance—a derangement, which preserves the mean budget by construction to four decimals—costs $+0.0011$ [$+0.0003, +0.0019$] (a second, independent derangement: $+0.0006, [-0.0004, +0.0016]$, straddling zero). The permutation is not inert: it displaces each instance’s budget by 1.44 ticks on average, against a between-instance standard deviation of 1.26. What the mixture *is* worth, relative to a flat uniform mixture at the same mean budget, is $+0.267$ [$+0.246, +0.288$]*—the learned population-level shape, not knowledge of the instance in front of it.* So the posterior neither knows nor acts: at matched readout and matched budget, knowing which instance needs how much compute is worth one part in a thousand.

Two readings follow. First, the hierarchy’s honest ceiling is the ex-ante rung: what the stake-class oracle can decide in advance is captured in full by the explicit allocator, and in this architecture self-observation adds essentially nothing beyond an artifact of how *our* adaptive-depth arms are read—a claim about PonderNet-style mixture readouts, untested in other adaptive-depth designs. Second, this retroactively explains *why* value on top of the posterior buys $+0.0002 \pm 0.0004$: we had read that null as elastic stopping already giving each instance what it needs, but the truth is plainer—per-instance allocation barely matters here at all, so there is nothing for value to improve. Value earns its keep only where allocation must be decided before observation is possible—a scope that the cliff experiment of §9 makes precise.

6 The precondition of governance: a frozen LLM actuator

Chapters above measure a substrate we trained. Modern practice, however, governs *frozen* language models at test time, most commonly by sampling n chains of thought and majority-voting (Wei et al., 2022b; Wang et al., 2023; Brown et al., 2024). We connected our instruments to a frozen Qwen2.5-14B-Instruct actuator (Qwen Team, 2024): tasks verbalized into text, stakes carried in metadata the actuator never sees, and every allocation policy evaluated offline over a single cached sample pool, so all arms are deterministic functions of the same generations.

The result reframes the question. Before asking *how* to govern test-time compute one must ask whether the lever has dynamic range, and here it has very little: real, measurably above zero, and bounded well below relevance. Across seventeen cells spanning two task families and base accuracies from 0.13 to 0.96, the probability that the modal answer of sixteen samples is wrong obeys an empirical law $P(\text{modal error}) \approx (1 - \text{acc}_1) - \epsilon$ with $\epsilon \leq 0.08$ in sixteen of the seventeen (the exception, the highest-slope cell, reaches 0.083): *the modal answer is essentially the answer of a single sample*, because errors are systematic—the model agrees with itself while being wrong (modal-error mass 0.833 in the frozen cell). Temperature partially decorrelates the errors, multiplying the vote-curve slope sevenfold and enlarging what lever exists—the *value-allocation ceiling*, the maximum payoff any stake-aware allocation of samples can gain over uniform at matched average samples, computed exactly from the measured vote curve with an unbiased without-replacement estimator, rises from about $+0.002$ at $T=0.7$ (a dial-round figure recorded only in the findings document and not re-derived with the corrected estimator) to $+0.0236$ at $T=1.3$ —but even then the ceiling is a two-sided bound: significantly above zero and significantly below our preregistered relevance threshold of 0.04 ($+0.0236$ [$+0.0150, +0.0326$], unbiased vote estimator, paired bootstrap). With that lever, the governance question does not arise—which is itself the finding: *the lever is a measurable property of the (actuator, spending mechanism) pair, and measuring it comes first.*

An anchor on a public benchmark. A natural objection is that self-consistency famously gains 10–18 points on GSM8K (Wang et al., 2023), which would put $\epsilon \approx 0.1$ –0.3, far above our law. At reviewer request we ran the anchor cell: 256 instances of the official GSM8K test with the same actuator and instruments ($m=16, T=0.7$). The law holds: $\text{acc}_1 = 0.905$, the vote curve peaks at 0.941 ($n=7$ –9, slope $+0.030$ to $n=15$), modal-error mass is 0.066—about 70% of the remaining error is systematic—and $\epsilon = +0.029 \leq 0.08$. There is no contradiction with the published gains: those were measured on much weaker samplers ($\text{acc}_1 \approx 0.4$ –0.6)

Dead hypothesis	Evidence	Measured mechanism
Field as compute source	$d_z = 4.06$ vs GRU	plan readout fails to extrapolate; inertia lags budget shifts
Field physics as load-bearing	$ \Delta \leq 0.012$ ($\times 3$ lines)	gradient coarse-grains time into set-points
Multiscale anticipation (S_5)	$\leq +0.001$ (31 configs)	smooth value; regimes mix faster than foresight pays
Routing via gradient	corr -0.01 (AUC 0.95)	state sheds task-irrelevant information
Consequences in the loss [†]	blind $+0.147$ better	ESS 0.40: weighting destroys the solver
Value atop the posterior	$+0.0002 \pm 0.0004$	per-instance allocation barely matters
Posterior allocation as information	$+0.0011$ [$+0.0003$, $+0.0019$] (readout fixed)	the mixture readout, not knowledge of the instance
The integration “touch”	-0.011 [-0.068 , $+0.045$]	unconfirmed; interval spans zero and the original
Field as evidence accumulator	$+0.0007$ [-0.0065 , $+0.0079$]	recurrence already integrates
Self-consistency as lever	ceiling $+0.0236 < 0.04$	systematic errors; modal ≈ 1 sample
LLM agreement as stopping	$\leq +0.017$ vs ex-ante (CI)	measures conviction, not correctness
Cliff family <i>constructible</i> on the small substrate (the prediction itself stays open there)	stuck at prior, 3 predeclared rounds	in-context retrieval blocked by the loop

Table 1: The negative space: where cognition does *not* live. All but one entry is a measurement with a mechanism rather than a bare null (the integration “touch” is a non-replication, reported as unresolved); rows are preregistered confirmatory results except the one marked [†], an exploratory branch the preregistration did not anticipate, and two audits run post hoc at reviewer request (posterior allocation; the LLM agreement intervals), all reported as such.

whose error mass was large and diversifiable, while a strong modern instruct model leaves little error and most of it systematic. The precondition finding—measure the lever’s range before optimizing over it—carries to the public benchmark unchanged.

The chapter’s sharpest result is a contrast of posteriors. On the trained substrate, self-observation *appeared* to be the greatest asset ($+0.223$) until the readout audit reduced it to $+0.0011$ at matched readout and budget. On the sampled LLM the analogous channel, inter-sample agreement, is nearly worthless too—and the two now agree. With paired bootstrap intervals computed at reviewer request, stopping on agreement buys $+0.017$ at most in point estimate (upper confidence limit $+0.025$) over ex-ante allocation ($+0.007$ [-0.008 , $+0.015$], $+0.015$ [$+0.001$, $+0.021$], $+0.017$ [$+0.005$, $+0.025$] at matched budgets of $\bar{n} \approx 3, 5, 7$ samples; an earlier point estimate had the sign negative, and the interval analysis corrects both the sign and the emphasis)—as close to worthless as the audited posterior advantage on the trained substrate, though an order of magnitude larger than it in absolute terms ($+0.017$ versus $+0.0011$)—while agreement mass concentrates on wrong answers (modal-error 0.833): agreement measures conviction rather than correctness, a failure mode distinct both from eliciting calibrated self-knowledge, which these models largely have (Kadavath et al., 2022; Lin et al., 2022), and from external verification (Cobbe et al., 2021; Lightman et al., 2024). Value on top of the LLM posterior is likewise null with intervals ($+0.003/+0.002/+0.001$, all straddling zero). The same act—watching oneself think—is nearly worthless in *both* regimes. That convergence was invisible while one of the two was measured through a readout the other did not share, which is the methodological point of this chapter.

7 Negative space as a map

A null without a mechanism is noise; with one, it is cartography. The program’s negative results, each with the measurement that explains it, are collected in Table 1. Together they draw the boundary that the positive results inhabit: the field does not think, the gradient does not route, the loss does not teach values, self-observation does not allocate, the sampled LLM’s agreement does not verify—and the places where each function actually lives are exactly the ones described in §3–§6.

8 Method as a result

The findings above are only as good as the process that produced them, and the process had to save us from ourselves repeatedly; we therefore report it as a contribution, in the spirit of the reproducibility literature (Nosek et al., 2018; Ioannidis, 2005; Simmons et al., 2011). Every confirmatory experiment was preregistered with complete verdict branches before data. Every design was attacked by an adversarial review panel before GPU was spent—across the program, seven panels produced more than two hundred findings, including genuine code bugs found by reviewers reading sources. Cheap kill-gates ran before expensive confirmatories, and calibration was firewalled at three declared dial rounds per family. Three episodes illustrate why none of this is ceremony.

Replication. The “touch” of §2 was significant at $p = 0.011$ and was not confirmed on replication—the replication interval covers both zero and the original effect—which is precisely why the replicas rule (two runs per cell for small effects) is now standing policy.

No null without a positive control. Twice in two days an instrument produced a false verdict in the null direction. The vote-curve estimator for the LLM chapter sampled with replacement from the cached pool, underestimating exactly where pools are diverse, and its ceiling was evaluated at a design cap rather than the affordable budget. Correcting both turned a ceiling once believed structurally unreachable into a real, measurable—but still sub-threshold—lever; a hasty re-measurement on 96 instances without a paired interval then produced a transient false positive (+0.050, above threshold)—our third error—before the full paired probe settled the honest two-sided bound of §6 (+0.0236, significantly above zero and significantly below 0.04). Out of the episode came a second standing rule: no verdict without an interval on the difference. Days later, the evidence-accumulator probe of §3 initially paid an overfitting tax that biased it toward null; a planted-signal control caught it before the verdict. The resulting rule—no null is reported unless its instrument demonstrably detects a planted effect of the threshold size—did the most work of any single practice in the program.

Pre-declared controls block bad instruments. An offline pilot of the cliff experiment burned three instrument designs in sequence; each was blocked by the same preregistered sanity condition (reproduce a known zero) before any conclusion could be contaminated, and the third failure revealed a structural fact about the known zero itself that redesigned the real experiment.

9 The cliff: executing our own prediction

Three independent chapters—anticipation, routing, and value atop the posterior—died against the same two properties of the smooth family: smooth value and an allocation problem in which which-instance-gets-what barely matters (§5). That convergence licensed a falsifiable prediction: in a *cliff* family, where task cost is all-or-nothing and invisible from inside, the hierarchy should tilt back toward ex-ante decisions. This section reports the execution of that prediction, in three acts.

Act one: the family resists construction on the small substrate. Our cliff family is *cycle walking*: given a random permutation presented as shuffled pairs, walk from s to t and report the distance d —unknowable until arrival, with only the cycle length L visible ex-ante. On the trained substrate the family failed its learnability gate three times, with a three-part mechanism. The task requires *in-context* retrieval (with the table in the weights, a spectral shortcut computes d in one pass and the cliff evaporates—the Fourier phenomenology of grokking (Power et al., 2022; Nanda et al., 2023)); the induction circuit that performs in-context retrieval (Olsson et al., 2022) forms only under a specific signal regime (dense supervision, large batch: loss sits at the exact support entropy $\ln 20$ for nine thousand steps, then transitions to perfect retrieval by fifteen thousand—on the plain backbone); and the recurrent loop that the family needs for its compute cliff *blocks that formation* (the same data regime that groks on the backbone stays at prior through thirty thousand steps under the reasoner). The component the family needs for the compute prevents learning the one it needs for the task. (The attribution is convergent rather than surgical: the round-3 run differs from the plain-backbone probe in one further respect—the distance-question head—though its auxiliary losses alone also stayed at $\ln 20$.) Signal regimes, like levers, are preconditions.

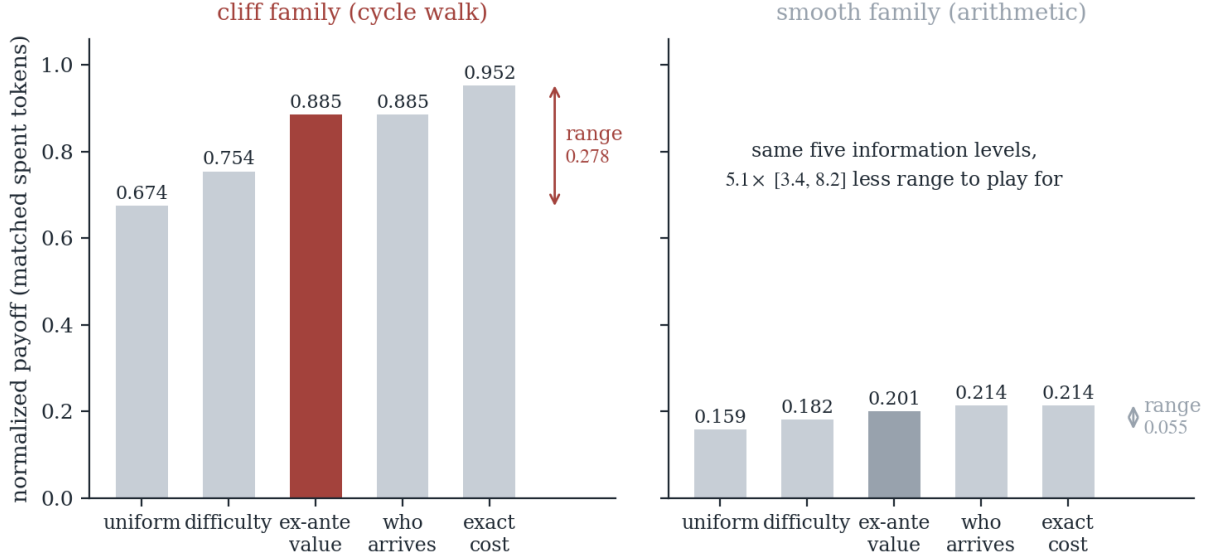


Figure 3: The cliff ladders on the frozen LLM actuator (left: cycle walking; right: arithmetic with visible work). All arms are offline truncations of the same recorded generations at matched spent tokens. The cliff multiplies the attainable allocation range fivefold (0.278 vs 0.055) with no *detected* change in the fraction of it that ex-ante information captures (0.759 vs 0.763), a comparison too imprecise to establish invariance. Bars are point estimates from single recorded greedy generations; the fraction comparison is post-hoc with a wide interval (difference -0.004 $[-0.150, +0.241]$), while the range ratio $5.1 \times [3.4, 8.2]$ excludes unity decisively (paired instance bootstrap, frozen allocations).

Act two: the family’s natural home. A pretrained LLM has induction heads for free, and its compute lever for this task is generation *length*, not votes: walking d steps costs tokens proportional to d (measured correlation 1.00), the visible class (the cycle length L , the only cost information available before walking) predicts cost poorly in the cliff family ($R^2 = 0.26$) and well in a smooth arithmetic family ($R^2 = 0.75$)—the operational definition of a cliff. The actuator genuinely walks: one hundred percent of correct answers contain a complete valid hop chain, so cost-proportionality holds literally. Every allocation arm is an offline truncation of a single recorded greedy generation per instance, re-parsed token-prefix by token-prefix, with budgets accounted as tokens *spent* and allocations solved exactly by dynamic programming.

Act three: the result. Figure 3 shows both ladders. In the cliff family, payoff climbs from 0.674 (uniform) through 0.754 (difficulty) to 0.885 (ex-ante value), and knowing *who will arrive* adds exactly nothing (0.885), while knowing the *exact cost* reaches 0.952. The preregistered headline confirms with room to spare: value minus difficulty is $+0.1312$ $[+0.1124, +0.1502]$, $3.4 \times$ its threshold, against $+0.019$ in the smooth family—value under commitment pays, and pays most where cost is invisible in advance. The preregistered opposite-direction contrast also confirms: the margin left above difficulty-only allocation is $+0.198$ in the cliff versus $+0.031$ in the smooth family.

And then the finding we did not seek. Normalizing each family by its own attainable range, the fraction captured by the best ex-ante policy is 0.759 $[0.703, 0.825]$ in the cliff and 0.763 $[0.522, 0.899]$ in the smooth family; the difference is -0.004 $[-0.150, +0.241]$ (paired instance bootstrap, frozen allocations)—a post-hoc observation compatible with equality, though the smooth family’s fraction is imprecisely estimated, so we claim no detected difference rather than invariance. The range ratio, by contrast, is solid: $5.1 \times [3.4, 8.2]$. The strong form of our prediction said the cliff would shift the information balance toward ex-ante decisions. It does not. *We detect no change in how the information is divided—though that comparison is imprecise—while the cliff demonstrably multiplies, fivefold, how much there is to divide* (0.278 versus 0.055). The prediction

conflated the magnitude of an allocation problem with the structure of its information; the data settle the magnitude axis and leave the structural one undecided.

Honesty requires two demotions. Our second hypothesis—that nothing realizable pays beyond arrival detection—*fails in its preregistered form*: exact-cost oracle minus arrival oracle is $+0.067$ against a 0.0056 threshold. We read that verdict as uninformative rather than damning—the preregistered comparator is clairvoyant rather than mid-flight, a flaw the design panel had flagged and we left standing, so the margin prices a clairvoyance no realizable observer has—but the formal verdict is a failure and we report it as one. What *is* realizable is measured and clean: the arrival oracle (a binary oracle of who finishes within budget) adds exactly nothing, and a redesigned mid-flight comparator is specified for future work. And strict muteness remains unadjudicated: the preregistered probe passed but its own positive control failed in its preregistered form, and under our no-null-without-a-control rule we do not claim it.

10 Discussion

The architecture that remains. Assembling the surviving positives yields a specific design. Competence is trained. Stopping is native and posterior—the halting head reading the evolving state—but the readout audit strips it of the role we had given it: its per-instance choices carry $+0.047$ when transplanted, less than difficulty information alone, and $+0.0011$ once the readout is equalized and the budget matched. The honest ceiling of the hierarchy is therefore the ex-ante rung, which the explicit allocator reaches to three decimals—an equality that is partly by construction, since our stake sensor is perfect. Value is computed by an explicit allocator over learned self-models—anticipation, in these families, has nothing to compute—and its domain is *commitment*: decisions that must be made before observation is possible, a domain the cliff result shows the cost structure of the environment can enlarge severalfold (range ratio $5.1 \times [3.4, 8.2]$ between our two families). The homeostatic field survives as what the companion paper called it: a certified compute governor—brainstem, not cortex (Anonymous, 2026)—and it survives as the *certifiable* implementation of that role rather than the best-performing one, since a GRU of matched interface governs at least as well.

One law, met three times. The program kept colliding with the same shape of fact at different levels. Before asking how to govern, measure whether the lever has range (§6). Before asking whether a family tests a hypothesis, measure whether its signal regime lets the substrate learn it at all (§9, act one). And when a capability is worth paying for, the payment scales with the ex-ante *invisibility* of cost, and not—so far as we could resolve it ($-0.004 [-0.150, +0.241]$)—with any change in who holds the information (§9, act three; a reading based on one pair of families and one actuator, offered as the sharpest available summary rather than an established law). Preconditions first; the interesting question is often one level below the one you asked.

Limitations. The trained substrate is small (4–6M parameters; 4.2–5.6M in the measured variants) and algorithmic by design; the hierarchy’s absolute numbers are properties of our task families, though the ordering of the three ex-ante rungs held in all twelve solvers (six seeds $\times$ two runs). The LLM chapters use a single frozen actuator; the governance chapter evaluates all arms offline over cached sample pools (768 instances, $m=16$ –24 samples per instance depending on the cell), while the cliff chapter uses greedy decoding with one recorded generation per instance. Replication across actuators (our preregistered universality hypothesis) was never run, though the GSM8K anchor cell extends the lever measurement to a public benchmark on the same actuator. Twelve replicas arise as six seeds $\times$ two runs, which is anti-conservative if runs share seed noise; we report per-seed sensitivity where it matters. Strict muteness and the realizable mid-flight ceiling remain open, with the redesigned comparator specified. Our stake is orthogonal to content by construction, and the measured mechanism of the computed-decision thesis is that training sheds task-irrelevant information from the state; with value correlated with content the verdict could differ, and we did not test it. The readout audit likewise concerns PonderNet-style halting whose language head is trained on the mixture, not every adaptive-depth design. And all governance results concern compute allocation; nothing here speaks to gradient-time governance.

11 Conclusion

We asked, of every part of a minimal complete cognitive architecture, whether its function emerges or must be computed, and obtained an answer with an unusual property: it is the same answer at every scale we probed. What can be observed turns out to be barely worth observing—the posterior halting head neither knows nor acts: forcing execution at the depths it chose reproduces its trajectory exactly (residual regime $+0.000$ $[0.000, 0.000]$), its per-instance choices are worth $+0.0011$ $[+0.0003, +0.0019]$ once the readout is equalized and the budget matched, and what looked like its advantage was the mixture readout ($+0.383$ $[+0.341, +0.424]$). What cannot be observed must be computed—explicitly, at decision time, from learned self-models, because gradients shed exactly the information those decisions need. And before either, the preconditions must be measured: the range of the lever, the regime of the signal, the visibility of the cost. The architecture of cognition, in this program’s experience, is less a stack of faculties than a discipline of measurement—and the map of where cognition does not live turned out to be the most reusable thing we built.

Reproducibility. All preregistrations (with dated amendments), findings documents, panel adjudications, solver checkpoints, caches, and analysis code are in the program repository. Numbers tied to a preregistered verdict enter the text as macros transcribed from the archived results and checked against them, with declared exceptions hard-coded from the same files (the H2 margin of §9, the cliff ladder rungs, and the agreement-stopping intervals of §6); the $T=0.7$ ceiling $+0.002$ and the exploratory $0.63/0.20$ pair are descriptive and carry no archived verdict file. Remaining descriptive figures are checked against the findings documents.

A Substrate, tasks, and payoff

Instances are length- K compositions over S_5 : the input lists K generator tokens (certified family: cycle generators), and the target at a fixed answer position is the composed permutation, with dense supervision of the running product during training. Out-of-distribution means K beyond the training range. Models are decoder-only transformers (4 layers, $d_{\text{model}}=256$, 4 heads, RMSNorm, RoPE, SwiGLU; 4–6M parameters depending on variant) feeding a recurrent reasoner block applied up to $N_{\text{max}}=24$ ticks under PonderNet-style halting, with an external slot memory. Training uses AdamW (cosine schedule, gradient clipping), BF16 on GPU with the physical field parameters pinned to FP32. Stakes are metadata: two slot tokens prepended to each instance, with stake high ($s=8$) iff the tokens are equal ($P=0.15$, independent of content and length by construction); the loss never sees them in the confirmatory arms. Payoff is stake-weighted accuracy $\sum_i s_i c_i / \sum_i s_i$ on a common frozen evaluation set of 16k instances. The twelve solvers are six seeds $\times$ two independent training runs of the stake-blind arm, evaluated frozen.

B Hierarchy arms and estimators

Ex-ante arms execute via per-instance forced depth with exact matching at $e=5$ mean ticks: uniform ($n=5$), difficulty (depth by class K from an isotonic success table $\hat{p}(\cdot | K)$ fitted on a probe split), rule (two levels by predicted stake, no success model), the explicit allocator ($n^* = \arg \max_n \hat{s} \hat{p}(\text{success} | n) - \lambda n$, with $\hat{s}$ a stake head read from slot embeddings and λ set by the budget), and the class oracle (true stake). The native arm is the trained halting run as trained. VG-B0 sweeps a stake-conditioned additive offset on the halting logit ($\delta \in \{0.5, 1, 2, 3.5\}$) at matched $\mathbb{E}[\bar{n}] \pm 0.05$. The yoked control re-executes every instance at forced depth round($\mathbb{E}[n_i]$) taken from the native rollout of that same instance. Contrasts are paired per instance and aggregated per solver ($n=12$; t -intervals), except the allocator contrast A1 of §4, which is aggregated per seed ($n=6$, $\text{df}=5$) and tested against the preregistered threshold $\delta_0=0.02$.

C LLM governance cells

The actuator is frozen Qwen2.5-14B-Instruct. Tasks are verbalized (S_5 -in-text: three named operations applied to a five-token row); stakes remain metadata carried outside the prompt. For each cell we cache a pool of m i.i.d. sampled chains per instance (768 instances, except the GSM8K anchor cell, which uses the

256 official test items; $m=16-24$; $T \in \{0.7, 1.0, 1.3\}$, top- $p=0.9$), and every policy is a deterministic function of the pool. The vote curve $\text{acc}(\text{vote-of-}n)$ uses the unbiased without-replacement estimator (subsets of the pool are i.i.d. draws by exchangeability; ties split uniformly; non-parses do not vote). The value-allocation ceiling is the maximum stake-aware gain over uniform at matched mean samples, computed exactly from the measured curve. Agreement-stopping policies are evaluated by dynamic programming over recorded sample-walk states (k , modal count), fitted and evaluated on disjoint halves; the paired bootstrap ($B=200$) resamples instances and repeats the full pipeline, including frontier construction.

D Cliff recording, truncation, and arms

Cycle-walk instances present a random single-cycle permutation over letters as shuffled pairs (cycle length $L \in \{6, 10, 14\}$ visible; distance $d \sim U[1, L-1]$ hidden); the arithmetic contrast family uses chained integer operations with the operation count visible. One greedy generation per instance is recorded as full token IDs. The arrival index c_i is the minimal token prefix containing a *completed* answer line (binary search over decoded prefixes), and $\text{correct}(\text{cap})$ re-parses the truncated prefix literally, which resolves multi-answer and split-digit artifacts (self-tested against literal re-parsing on a cap grid). Budgets are accounted as tokens *spent*, $\sum_i \min(t_i, \text{cap}_i)$, matched across arms; allocations are solved exactly by multi-choice knapsack dynamic programming over a cap grid; payoffs marginalize the stake analytically; caps are cross-fitted ($A \leftrightarrow B$ halves). Gates run on the recordings before any hypothesis is computed: R (cost-distance correlation 1.00; cost predictability from the visible class R^2 0.26 vs 0.75), S (100% of correct answers contain a complete valid hop chain), and censoring (0.4%, 0.4%, 0.0% at $L=6, 10, 14$). The fraction and range-ratio intervals use a frozen-allocation instance bootstrap ($B=2000$) with the exact-cost oracle re-solved per replicate.

Author contributions (CRediT). Withheld for double-blind review.

Data and code availability. The full program—model code, preregistrations, run artifacts and the scripts that produce every table and figure in this paper—is covered by the MIT licence. The repository is withheld here for double-blind review; its URL and an archival DOI will be given in the camera-ready version. Every number in the text is a macro bound to an archived artifact, so the manuscript cannot drift from the data.

Use of generative AI. The experiments were designed, executed and adjudicated by the authors. A large language model was used as a coding and drafting assistant throughout, including for implementation, for adversarial review of the manuscript’s internal consistency, and for literature search; all claims, numbers and verdicts reported here were verified by the authors against the archived artifacts.

References

- John R. Anderson, Daniel Bothell, Michael D. Byrne, Scott Douglass, Christian Lebiere, and Yulin Qin. An integrated theory of the mind. *Psychological Review*, 111(4):1036–1060, 2004.
- Cem Anil, Yuhuai Wu, Anders Andreassen, Aitor Lewkowycz, Vedant Misra, Vinay Ramasesh, Ambrose Slone, Guy Gur-Ari, Ethan Dyer, and Behnam Neyshabur. Exploring length generalization in large language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2207.04901.
- Anonymous. Can a dynamic internal field govern a transformer’s cognition? certifiability, not superiority, in homeostatic compute control. *Companion submission under review*, 2026.
- Andrea Banino, Jan Balaguer, and Charles Blundell. PonderNet: Learning to ponder. In *8th ICML Workshop on Automated Machine Learning (AutoML)*, 2021. arXiv:2107.05407.
- Lisa Feldman Barrett. The theory of constructed emotion: An active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience*, 12(1):1–23, 2017.
- Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*, 2024.

- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. Universal transformers. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1807.03819.
- Maha Elbayad, Jiatao Gu, Edouard Grave, and Michael Auli. Depth-adaptive transformer. In *International Conference on Learning Representations (ICLR)*, 2020.
- Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
- John C. Gittins. Bandit processes and dynamic allocation indices. *Journal of the Royal Statistical Society, Series B*, 41(2):148–177, 1979.
- Alex Graves. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016.
- Nicholas Hay, Stuart Russell, David Tolpin, and Solomon Eyal Shimony. Selecting computations: Theory and applications. In *Uncertainty in Artificial Intelligence (UAI)*, 2012.
- Eric J. Horvitz. Reasoning about beliefs and actions under computational resource constraints. In *Proceedings of the Third Workshop on Uncertainty in Artificial Intelligence*, 1987.
- John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005.
- Saurav Kadavath, Tom Conerly, Amanda Askell, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- John E. Laird. *The Soar Cognitive Architecture*. MIT Press, 2012.
- Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.
- Yann LeCun. A path towards autonomous machine intelligence. *OpenReview preprint*, 2022.
- Falk Lieder and Thomas L. Griffiths. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences*, 43:e1, 2020.
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, et al. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022.
- Bingbin Liu, Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. Transformers learn shortcuts to automata. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.10749.
- Aman Madaan, Niket Tandon, Prakhar Gupta, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Kingson Man, Antonio Damasio, and Hartmut Neven. Need is all you need: Homeostatic neural networks adapt to concept shift. *arXiv preprint arXiv:2205.08645*, 2022.
- William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics (TACL)*, 11:531–545, 2023.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.

- Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- David Raposo, Sam Ritter, Blake Richards, Timothy Lillicrap, Peter Conway Humphreys, and Adam Santoro. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*, 2024.
- Stuart Russell and Eric Wefald. Principles of metareasoning. *Artificial Intelligence*, 49(1-3):361–395, 1991.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Roy Schwartz, Gabriel Stanovsky, Swabha Swayamdipta, Jesse Dodge, and Noah A. Smith. The right tool for the job: Matching model and instance complexities. In *Association for Computational Linguistics (ACL)*, 2020.
- Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011.
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- Nicolas Vecoven, Damien Ernst, Antoine Wehenkel, and Guillaume Drion. Introducing neuromodulation in deep neural networks to learn adaptive behaviours. *PLOS ONE*, 15(1):e0227922, 2020.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- Jason Wei, Yi Tay, Rishi Bommasani, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022a.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022b.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Hattie Zhou, Arwen Bradley, Etai Littwin, et al. What algorithms can transformers learn? a study in length generalization. In *International Conference on Learning Representations (ICLR)*, 2024.